

Some Mixed-Signal Oscilloscopes in the Market

Manufacturer	Model/type	Features	Price
Bitscope	BS05, PC based	• 20MHz bandwidth, two analogue and six digital channels • Compatible with Windows, Linux, Mac OS X and Raspberry Pi • Decodes Serial, SPI, I2C and CAN	₹ 14,131*
CWAV Inc.	USBee QX, PC based	• First-of-its-kind USB 3.0/Wi-Fi MSO and protocol analyser • Captures up to 32 days of traces • 600MHz with 24 digital channels and four analogue channels • Includes popular serial protocols such as RS232/UARTs, SPI, I2C, CAN, SDIO, Async, 1-Wire and I2S	US\$ 2495
Keysight Technologies	InfiniiVision 3000T X-Series, benchtop	• 21.6cm (8.5-inch) capacitive touch screen, waveform update rate up to 1M wfms/s, six-in-one integration that includes an oscilloscope, optional digital channels, hardware serial protocol decode and triggering, built-in arbitrary waveform generator, DVM and an 8-digit counter and totaliser	Base price ranging between ₹ 340,278 and ₹ 1,080,832
Oscium	iMSO-204L, mobile/tablet based	• MSO designed specifically for iPhones, iPads, iPods • Two analogue and four digital channels, 50MS/s sample rate, 5MHz bandwidth, 200ns/div-10s/div	Starts from ₹ 38,000**
Rigol	MSO1000 series/ MSO2000 series/ MSO4000 series, benchtops	• 70MHz-500MHz models with 16 digital channels and two or four analogue, up to 4Gsa/s sampling, up to 140Mpts record length, serial triggering and decode, inbuilt 2-channels 25MHz arbitrary function generator, 6-bit counter	Starts from US\$ 835/US\$ 1239/US\$ 3217
Siglent	SDS2000 series, benchtop	• Up to 300MHz bandwidth, 2Gsa/s sampling rate, built-in 25MHz function/arbitrary waveform generator and a variety of serial protocol trigger and decode functions	Ranging between US\$ 805 and US\$ 2558
Tektronix	MSO4000B series, benchtop	• Up to 1GHz bandwidth and 5GS/s sample rate, digital phosphor display and up to four analogue and 16 digital channels	Ranging between ₹ 1,020,000 and ₹ 1,570,000
Teledyne LeCroy	HDO8000 series, benchtop	• 8-channel available in 350MHz, 500MHz and 1GHz bandwidths • HD4096 technology and ERES (enhanced resolution) filtering	Starts from US\$ 23,850
Yokogawa	DLM4000 series, benchtop	• Available in both 350MHz and 500MHz models; 2.5Gsa/sec DLM4000 features eight analogue input channels • Features Flex MSO • High-resolution 30.7cm (12.1-inch) LCD display	N/A

*As listed on element14.com

**As notified by the Indian distributor

Engineers are always on the look out for more analogue channels in a scope. Eight analogue channels provide great flexibility to an engineer working on complex embedded designs. Although, more analogue channels on an MSO makes the device more expensive. Apart from Teledyne LeCroy's HDO8000 series of MSOs, another test and measurement (T&M) vendor, Yokogawa, provides eight analogue channels in its DLM4000 series of MSOs.

Such MSOs, with a wide selection of channels, are ideal for professional debugging high-power power electronics, automotive electronics and complex embedded as well as mechatronic designs. A unique ability of DLM400 series is that the eighth analogue channel comes with Flex MSO, a feature that converts it to an

8-bit logic analyser with the push of a button.

With more analogue channels and higher resolutions, deep record lengths is an equally important feature. T&M vendors are offering MSOs with deep memory typically ranging from 150Mpts per channel to 250Mpts per channel. The sampling rate for MSOs usually range from 1.5Gsa/s to 4Gsa/s.

CAN FD-enabled MSOs

Most of the known MSO vendors are steadily incorporating support for controller area network with flexible data rate (CAN FD) data frames by providing an additional trigger capability for better analysis of traffic in the CAN FD network.

Apart from Teledyne LeCroy, Keysight and Rohde & Schwarz, Yokogawa

has integrated support for CAN FD in their DLM2000 and DLM4000 series of MSOs. Evolution of more complex and advanced vehicular control systems has a proportional effect on the amount of traffic inside the CAN network. Therefore it is important for MSOs to have the ability to provide engineers with dedicated triggers and functionalities to boost the analysis and debugging process.

Apart from triggers and search functions for CAN FD, it is important for the MSO to have high memory depth as it makes it easier for the designer to handle extended data frames.

Some other oscilloscopes that support CAN FD are Infinii Vision 3000T X-series and 4000T X-series from Keysight Technologies, Teledyne